SUMMARY INFOGRAPHIC · IAEA TRS-398 · EDUCATION REFERENCE

Reference Dosimetry, Explained

How a small electric charge from an ion chamber becomes a trusted absorbed dose, in one page.

01 The big idea

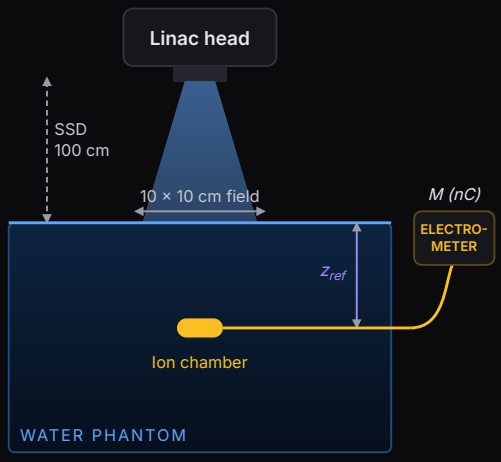
An **ionization chamber** is a small pocket of air. Radiation knocks electrons off the air molecules and the **electrometer** collects that charge as a reading, M, in nanocoulombs.

Charge is not dose. A standards lab has measured how many **grays per nanocoulomb** your chamber gives in cobalt-60: the calibration coefficient N_{D,w,Q_o}.

You **measure under standard conditions**, correct the reading for today's air and chamber behaviour, and convert it to your accelerator's beam.

$$D_{w,Q} = M_Q \times N_{D,w,Q_o} \times k_{Q,Q_o}$$

02 The measurement setup



Photons: z_{ref} = 10 g/cm², SSD or SAD 100 cm.
Electrons: z_{ref} = 0.6 R_{50} − 0.1 g/cm².

03 From reading to dose: a worked example (worksheet default photon inputs, NE 2571 chamber)

RAW READING	CORRECTIONS	CORRECTED	CONVERT	DOSE TO WATER
M_1	× k-factors	M_Q	× $N_{D,w}$ × k_Q	$D_{w,Q}$
1.950 nC	**× 1.0134**	**1.976 nC**	**× 0.0521 × 0.998**	**0.1028 Gy**
What the electrometer shows.	k_{TP}·k_{elec}·k_{pol}·k_s combined.	As if at calibration conditions.	Gy/nC in ^{60}Co, adjusted to this beam.	Dose at z_{ref} for that irradiation.

04 Each correction, explained

	FACTOR	WHAT GOES WRONG	HOW IT IS FIXED	EXAMPLE
k_{TP}	**Air density**	Warmer air or lower pressure means less air in the cavity: reading drops.	Scale by (273.2+T)/(273.2+T_0) × P_0/P.	1.0094
k_{pol}	**Polarity**	Reading changes slightly between + and − collecting voltage.	Measure both: ($\|M_+\|$ + $\|M_-\|$) / 2M.	1.0000
k_s	**Recombination**	Some ions recombine before collection: charge is lost.	Two-voltage method: M_1 at V_1 vs M_2 at lower V_2.	1.0040
k_{elec}	**Electrometer**	The electrometer itself may read slightly off.	From the certificate; 1.000 if calibrated with the chamber.	1.0000
k_{Q,Q_o}	**Beam quality**	Chamber was calibrated in ^{60}Co, not in your beam.	Table lookup by chamber model and $TPR_{20,10}$ or R_{50}.	0.9980

05 Photons vs electrons

PHOTON BEAMS
Quality: $TPR_{20,10}$
Depth: 10 g/cm²
Field: 10 × 10 cm
Chamber: cylindrical

ELECTRON BEAMS
Quality: R_{50}
Depth: 0.6 R_{50} − 0.1 g/cm²
Field: ≥ 10 × 10 cm²
Chamber: plane-parallel

06 Key takeaways

1. **Standard conditions** make every calibration comparable.
2. **Every factor should be close to 1.** A large one points to a setup or equipment problem.
3. **Traceability** runs from your result back to a primary standard.
4. **Record everything**: the worksheet is your audit trail.

Teaching aid only. Follow the full IAEA TRS-398 text and local protocols for clinical calibration.